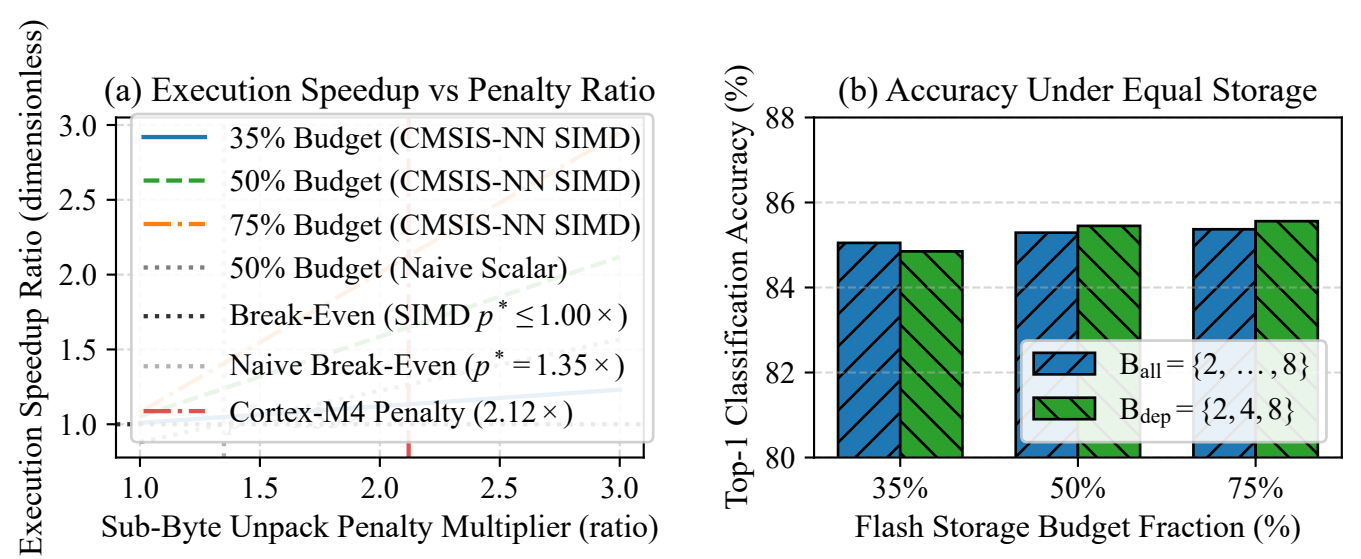
(a) Execution Speedup vs Penalty Ratio
Execution Speedup Ratio (dimensionless)
Sub-Byte Unpack Penalty Multiplier (ratio)
35% Budget (CMSIS-NN SIMD)
50% Budget (CMSIS-NN SIMD)
75% Budget (CMSIS-NN SIMD)
50% Budget (Naive Scalar)
Break-Even (SIMD $p^* \leq 1.00\times$)
Naive Break-Even ($p^* = 1.35\times$)
Cortex-M4 Penalty (2.12×)
3.0
2.5
2.0
1.5
1.0
1.0
1.5
2.0
2.5
3.0
(b) Accuracy Under Equal Storage
Top-1 Classification Accuracy (%)
Flash Storage Budget Fraction (%)
$B_{all} = \{2, \ldots, 8\}$
$B_{dep} = \{2, 4, 8\}$
88
86
84
82
80
35%
50%
75%